

Research Associate at Dartmouth College | Polar Engineer/Glaciologist
derek@dartmouth.edu | www.dereckpickell.com

EDUCATION

Doctor of Philosophy

Aug 2021 - Sep 2025

Dartmouth College, Hanover, NH

Thesis: *Old, Flat, and Slow: Interior Greenland Snow and Ice Dynamics Revealed with GNSS*

Hardware and software designs of low-power, scientific-grade GNSS: github.com/glaciology/OGRE

Advisor: Dr. Robert L. Hawley

- Lead for various GNSS field projects, including validating NASA laser altimetry measurements, studying crevasse formation, improving firn densification models, and improving space-based ice velocity estimates.
- Designed OGRE (Open GNSS Research Equipment), a field-deployable GNSS instrument for polar research, reducing costs by 44-90% compared to commercial alternatives
- Optimized OGRE for extreme environments, achieving best-in-class low power consumption, minimizing battery requirements for long-term, over-winter autonomous deployments in remote field settings.
- Developed an autonomous radar technique for monitoring snow surface roughness, enhancing the temporal and spatial resolution of data in understudied polar regions.
- Deployed and supported over 100 OGRE units across Greenland, the USA, and Antarctica as part of 7+ funded research projects, supporting studies in geophysics, hydrology, crevassing, and ice sheet dynamics.
- Accumulated over 52 weeks of field experience in Arctic and Antarctic environments, demonstrating expertise in remote field operations, instrument deployment, and extreme environment logistics.

Bachelor of Arts in Physics - Applied Track

June 2020

Pomona College, Claremont, CA

Thesis: *Computer Vision Software Development for an Explosive Seed Pod Trap Camera*

Engineering Capstone: *Position Determination from Star Tracker Earth Observations*

University of Cambridge, Cambridge U.K. (2019): *Characterizing Chemical Abundances of Select A-type Stars*

PUBLISHED PAPERS

- [5] **Pickell, D. J.**, Hawley, R. L., Good, J. C. ICESat-2 surface elevation assessment with kinematic GPS and static GNSS near the ice divide in Greenland. *The Cryosphere*. 2026.
- [4] **Pickell, D. J.**, Hawley, R. L., Lewinter, A. Spatiotemporal patterns of accumulation and surface roughness in interior Greenland with a GNSS-IR network. *The Cryosphere*. 2025.
- [3] Wilner, J. A., Nordin, B. J., Getraer, A., Gregoire, R. M., Krishna, M., Li, J., **Pickell, D. J.**, Rogers, E. R., McDannell, K. T., Palucis, M. C., Keller, C. B. Limits to timescale dependence in erosion rates: Quantifying glacial and fluvial erosion across timescales. *Science Advances*. 2024.
- [2] Khan, S. A., Morlighem, M., Ehrenfeucht, S., Seroussi, H., Choi, Y., Rignot, E., Humbert, A., **Pickell, D. J.**, Hassan, J. Inland summer speedup at Zachariæ Isstrøm, northeast Greenland, driven by subglacial hydrology. *Geophysical Research Letters*. 2024.
- [1] **Pickell, D. J.**, Hawley, R. L. Performance characterization of a new, low-cost multi-GNSS instrument for the cryosphere. *Journal of Glaciology*. 2024.

PAPERS IN REVIEW & MANUSCRIPTS

- Submitted March 2026. **Pickell, D. J.**, Hawley, R. L., Stevens, C. M. A single instrument approach to ice flow, surface mass balance, and firn densification validation in central Greenland.

- *In Review April 2026.* Sledd, A., Gallagher, M. R., Shupe, M. D., Cox, C. J., Hawley, R. L., Town, M. S., Guy, H., Marshall, H.-P., Neely, R. R., III, Pettersen, C., Walden, V. P., Hebson, C., Martin, A., Olson, E., **Pickell, D. J.** Summer subsurface temperature variability in the percolation zone of southwest Greenland: high resolution observations of the top meter of firn. *EGU sphere* [preprint]. <https://doi.org/10.5194/egusphere-2026-1842>. 2026.
- *In Review June 2026.* Gallagher, M. R., Shupe, M. D., Walden, V. P., Sledd, A., Cox, C. J., Guy, H., Pettersen, C., Martin, A. S., Neely, R. R., III, Hawley, R. L., Town, M. S., Olson, E., Marshall, H.-P., **Pickell, D. J.**, Vandecrux, B., Mottram, R., Mattingly, K. S., Clemens-Sewall, D., Hebson, C. How Greenland melts under cloudy and clear skies. Planned submission: April 2026.
- *In preparation.* **Pickell, D. J.**, Scott, J. GNSS interferometric reflectometry reveals detailed lake surface phenology. Planned submission: mid 2026.
- *In preparation.* **Pickell, D. J.** Widespread changes in Greenland surface roughness influences how it melts (with ICESat-2). Planned submission: mid 2026.

CONFERENCE ABSTRACTS, PRESENTATIONS, & INVITED TALKS

- [12] **Pickell, D. J.** ICESat-2 surface elevation assessment with kinematic GPS and static GNSS near the ice divide in Greenland. NASA ICESat-2 Science Project Office talk. 2026.
- [11] Togaibekov, A., Khan, S. A., Løkkegaard, A., Colgan, W., **Pickell, D. J.** Temporal evolution and kinematics of a mega-crevasse at Jakobshavn Isbræ revealed by dense GNSS observations. *EGU General Assembly 2026*.
- [10] **Pickell, D. J.**, Hawley, R. L., Good, J. C. ICESat-2 surface elevation assessment with kinematic GPS and static GNSS near the ice divide in Greenland. *The Cryosphere*. 2026.
- [9] Wilner, J. A., Nordin, B. J., Getraer, A., Gregoire, R. M., Krishna, M., Li, J., **Pickell, D. J.**, Rogers, E. R., McDannell, K. T., Palucis, M. C., Keller, C. B. Global quantification of glacial versus fluvial erosion rates: limits to timescale dependence. *EGU General Assembly 2025*.
- [8] Løkkegaard, A., Colgan, W., Khan, S. A., Fahrner, D. R., Polzin, M., Hughes, J., Lippert, E. Y., **Pickell, D. J.** Brittle landscapes: A case study of crevasse development in firn. *EGU General Assembly 2025*.
- [7] **Pickell, D. J.**, Hawley, R. L. Spatio-temporal patterns of surface accumulation and surface roughness in the interior region of Greenland with a novel GNSS network. *AGU Fall Meeting 2024*.
- [6] Gallagher, M. R., Walden, V. P., Shupe, M. D., Guy, H., Martin, A. S., Sledd, A., Pettersen, C., Neely, R. R., III, Marshall, H.-P., Hawley, R. L., Olson, E., Mattingly, K. S., Hebson, C., Neff, W. D., **Pickell, D. J.**, Town, M. S. How Greenland melts: high fidelity observations of the coupled climate system in Southwestern Greenland from summer 2024. *AGU Fall Meeting 2024*.
- [5] **Pickell, D. J.**, Hawley, R. L. Surface mass balance and elevation change in Greenland with a network of low-cost GNSS stations. *AGU Fall Meeting 2023*.
- [4] Good, J. C., Hawley, R. L., **Pickell, D. J.** Advancing snow surface roughness observations of the Greenland Ice Sheet using down-looking laser distance meters. *AGU Fall Meeting 2023*.
- [3] **Pickell, D. J.**, Hawley, R. L. Altimetry validation and snow surface monitoring with an on-ice array of open source multi-GNSS instruments. *AGU Fall Meeting 2022*.
- [2] **Pickell, D. J.**, Hawley, R. L. An on-ice GNSS Research Experimental Network for Greenland (OGRENet). *Northeastern Glaciology Meeting 2021*.
- [1] **Pickell, D. J.**, Hawley, R. L. An on-ice GNSS Research Experimental Network for Greenland (OGRENet). *AGU Fall Meeting 2021*.

PUBLISHED DATASETS

- [4] **Pickell, D. J.**, Hawley, R. L., Brunt, K. M., Dorsi, S. W., McConnell, J. R., Neumann, T. A. & Pettit, J. ICESat/ICESat-2 Traverse: Monthly GPS Surface Elevation Data at Summit Station, Greenland. (IS2TGPSSS, Version 1). [Data Set]. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. 2026.

- [3] Hawley, R. L., **Pickell, D. J.**, McConnell, J. R., Neumann, T. A., Felikson, D. & Dorsi, S. W. Weekly snow stake array measurements, Summit Station, Greenland (IS2SSAGCV, Version 1). Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. 2026.
- [2] **Pickell, D. J.**, Hawley, Robert L., Brunt, Kelly M., Burkhart, John F., Dorsi, Samuel W., McConnell, Joseph R., Neumann, Thomas A., Pettit, Joe. ICESat/ICESat-2 traverse: monthly GPS surface elevation data at Summit Station, Greenland. Dartmouth Dataverse 2025.
- [1] **Pickell, D. J.**, R. L. Hawley. An East-West Network of Twelve Global Navigation Satellite System (GNSS) Stations in the Summit Region of Greenland: RINEX Data, 2022-2025. Arctic Data Center. 2024.

SIGNIFICANT FIELD EXPERIENCE

- Drang Drung Glacier, India, *Expedition Member*, ground radar (radio echo), 1 week 2026
- McMurdo/Ross Ice Shelf Shear Zone, *Expedition Member*, ground penetrating radar, 4 weeks 2025
- Qaanaaq, Greenland, *Expedition Member*, GNSS instrument deployment, 3 weeks 2025
- Summit, Greenland, *Expedition Lead*, snow and ice dynamics, 2 weeks 2025
- Athabasca Glacier, Canada, *Team Instructor/Leader*, glaciology field education program, 1 week 2022, 2024, 2025
- Leverett Glacier, Antarctica, *Radar Engineer + Mountaineer Safety*, Crevasse mapping, 4 months 2024
- Mega-crevasse Zone, Ilulissat, Greenland, *Expedition Member*, GNSS instrument deployment, 2 weeks 2024
- Point 660, Greenland, *Educational Lead*, ice dynamics and mapping, 3 weeks 2024
- Summit, Greenland, *Expedition Lead*, snow and ice dynamics, 2 weeks 2024
- Summit, Greenland, *Science Technician*, meteorology, geodesy, glaciology, 9 months 2021, 2023
- North East Greenland Ice Stream, Greenland, *Expedition Member*, ice dynamics, 1 week 2023
- Summit, Greenland, *Expedition Lead*, snow and ice dynamics, 2 weeks 2023
- Summit, Greenland, *Expedition Member*, surface processes/instrument deployment, 2 weeks 2022
- Wilderness First Responder training 2016, 2022, 2025

TEACHING, MENTORSHIP & OUTREACH

- Leh Indian Glaciology Summer School - Guest Lecturer Summer 2026
- EARS46 - Earth Science Field Study Program, Athabasca Glacier Fall 2024
 - Led GNSS, glacier flow, and glacier travel/safety exercises in the Columbia Ice Field
- Joint Science & Education Project Graduate Teaching Fellow, Greenland Summer 2024
 - 3 week Arctic field school in Greenland for Danish, American and Greenlandic high school students
 - Taught basic glacier dynamics and mass balance module and supervised student project on radar and GNSS ice measurements
- 'Geolunch' Weekly Science Seminar Coordinator Fall 2022
- EARS46 - Earth Science Field Study Program Spring 2022
- EARS01 - How the Earth Works Fall 2021

MENTORSHIP & ADVISING

- Evan Barrett, *Thesis Committee Member*
- Maia Crichlow, *Advisor, Senior Capstone Project*
- Jacob Gordon, *Advisor, Senior Capstone Project*
- Laura Wilson, *Mentor/Field Lead, Summit Greenland*
- Jadin Scott, *Advisor, Research Assistant*

FUNDING, GRANTS & AWARDS

Pending Support

- Submitted March 2026 (PI). *Study of Height, Accumulation, Roughness, and Densification at Summit (SHARDS)*. NASA ICESat-2 Project Office. \$150,680
- Submitted March 2026 (Co-PI). *Geophysical Network Of MEMS for Englacial Sensing (GNOMES)*. Arete Glacier Initiative Foundation. \$191,622.88

Funded Support

- Evans Family Cluster Funding (Dartmouth College): *OGREs for Open GNSS* (2025) \$23,000.00
- Navigating the New Arctic (NNA) Arctic Observing Technologies Research Award (2025) \$1690.00
- Joint Science and Education Program Fellowship (2024) \$4000.00
- The Arthur L. Irving Institute for Energy & Society Independent Research Grant (2024) \$3000.00
- Dartmouth College Guarini Graduate Fellowship (2021 - 2025) \$88,164.00/yr

Other Funding/Awards

- Earle S. Lenker 1956 Graduate Award for outstanding academic achievement by a graduate student 2026

CONSULTING

- Summit Station Science Coordination Office (SCO) 2024 - present
 - Member of 5 person SCO, which advises the National Science Foundation and Arctic logistics contractors on various ongoing projects in Greenland in which science operations and research is a stakeholder.
- Faculty Search Committee Fall 2024
- Reviewer, *Journal of Glaciology*, *IEEE Transactions on Geoscience and Remote Sensing*

WORK EXPERIENCE

Greenland Science Technician, Polar Field Services Jan 2023 - Jun 2023; Apr 2021 - Jun 2021

- Ensured continuous operation of 15+ scientific instruments in an extreme polar environment, enabling uninterrupted atmospheric and geophysical monitoring for global research initiatives.
- Led field operations for 7 NASA ground data collection traverses to validate the NASA Ice, Cloud, and land Elevation Satellite (ICESat-2), directly contributing to improved satellite-based ice sheet mass balance models.
- Upgraded UNAVCO GPS infrastructure, improving precision of geodetic measurements used for tectonic studies and glacier movement tracking.
- Developed and executed action plans to support field research teams and maximize ground-time efficiency, including radio neutrino installation, ice core drilling, and space probe testing.

Loon (Google X) Flight Assembly Specialist, Polar Field Services August 2020 - February 2021

- Carried out assembly, integration, software/electrical testing and quality control of Loon flight systems at field operations facilities, leading to 30+ balloon launches in a three month period.
- Wrote test plans, performed outgoing preflight certification checks and executed launches with launch team.
- Triaged flight system test failures with engineering team while documenting bugs and process improvements.

Research Engineer: Millennium Space Systems September 2019 - May 2020

- Led a technical trade study on optical-based positioning methods as part of a Harvey Mudd College Clinic team, identifying system improvements to enhance satellite navigation and autonomy.
- Developed and implemented image-processing algorithms and software that expanded the satellite star tracker's capabilities, enabling multi-source navigation using Earth features, increasing operational time of star trackers to 100% when not viewing the sun.

Structural & Field Engineer Intern: Kiewit May - August 2018, June - August 2019

- Designed supporting structures for large-scale infrastructure projects in Seattle and Los Angeles, utilizing 3D modeling software to optimize designs for cost, efficiency, and constructibility.
- Led the transition to a streamlined CAD review system, improving design accuracy and reducing time-to-build by 1-2 weeks, contributing to faster project approvals and cost control.
- Performed structural calculations and load analyses to assess new and existing formwork, ensuring compliance with engineering standards and safety regulations.
- Designed critical safety and mechanical systems, including fall protection and formwork, enhancing worker safety and structural integrity on active construction sites.
- Assisted in managing a 20+ person construction crew by developing engineering-based work plans and hands-on training programs, ensuring successful implementation of design modifications in the field.

TECHNICAL SKILLS

- **Electronics:** PCB manufacturing & DC circuit design (Eagle CAD), bench top debugging (oscilloscope, multimeter).
- **Programming/Software:** Python, C/C++, bash/CLI. Focus on scientific/data analysis scripting, embedded firmware software, cloud computing (e.g., SlideRule), and software-defined radio (SDR) radar.
- **Spatial analysis:** GNSS reflectometry, geostatistics (kriging, noise modeling, spatiotemporal analysis), GNSS Field Surveying/Data processing (RTK, PPK, PPP), ArcGIS/QGIS (data processing and visualization), ENVI (remote geospatial data processing).
- **Mathematical/analytical tools:** Matlab, Mathematica, Excel, LabView.
- **3D modeling:** Autodesk, SketchUp and Microstation.
- **Media processing & creation:** Machine Shop (CNC router, 3D printing, lathe, etc.), Advanced proficiency in Adobe Suite.
- **Piloting:** 120+ hours as private pilot, tailwheel and seaplane endorsements.
- **Foreign languages:** Advanced in Spanish, proficient in Portuguese, elementary Mandarin.

ACTIVITIES

- **Outdoor education/exploration:** Wilderness First Responder, Pomona College outdoors club trip leader. Selected summits include El Capitan (via The Nose) and Mt. Rainier (Emmons-Winthrop Route, DC Glacier Route).
- **Vans RV-7 aircraft build:** Assisted the building of an experimental class aircraft with Harvey Mudd College students.